Lab Report No 5

Digital Signal Processing

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Section: B

Student Signature:

Department of Computer Systems Engineering

University of Engineering and Technology Peshawar

CSE 402L: Digital Signal Processing

Demonstration of Concepts	**Poor (Does not meet expectation (1))** The student failed to demonstrate a clear understanding of the assignment concepts	**Fair (Meet Expectation (2-3))** The student demonstrated a clear understanding of some of the assignment concepts	**Good (Exceeds Expectation (4-5)** The student demonstrated a clear understanding of the assignment concepts	**Score** **30%**
Accuracy	The student completed (<50%) tasks and provided MATLAB code and/or Simulink models with errors. Outputs shown are not correct in form of graphs (no labels) and/or tables along with incorrect analysis or remarks.	The student completed partial tasks (50% - <90%) with accurate MATLAB code and/or Simulink models. Correct outputs are shown in form of graphs (without labels) and/or tables along with correct analysis or remarks.	The student completed all required tasks (90%-100%) with accurate MATLAB code and/or Simulink models. Correct outputs are shown in form of labeled graphs and/or tables along with correct analysis or remarks.	**30%**
Following Directions	The student clearly failed to follow the verbal and written instructions to successfully complete the lab	The student failed to follow the some of the verbal and written instructions to successfully complete all requirements of the lab	The student followed the verbal and written instructions to successfully complete requirements of the lab	**20%**
Time Utilization	The student failed to complete even part of the lab in the allotted amount of time	The student failed to complete the entire lab in the allotted amount of time	The student completed the lab in its entirety in the allotted amount of time	**20%**

Lab No: 5 .

Title: Analysis of Amplitude Modulated and Demodulated Signal using Matlab

Provide .m file with detailed comments

Tasks:

1. **Define Amplitude Modulation:**
   Amplitude Modulation is the process of varying the amplitude of a carrier signal in proportion to the modulating signal while maintaining the carrier's frequency constant.

2. **Define Amplitude Demodulation:**
   Amplitude Demodulation is the process of extracting the original message signal from an amplitude-modulated (AM) carrier signal.

3. **List three reasons, why we implement Amplitude Modulation in Communication Systems**

- Efficient Transmission of Signals Over Long Distances
- Smaller Antenna Size
- Maximum Bandwidth utilization

4. **Define Modulation Index**
   Modulation index is defined as the ratio of the peak amplitude of the message signal to the peak amplitude of the carrier signal.

5. **Input Modulation Index from 0 to 1.4, the increment step should be 0.2. Observe/analyze and comment about the output observed.**

Procedure:

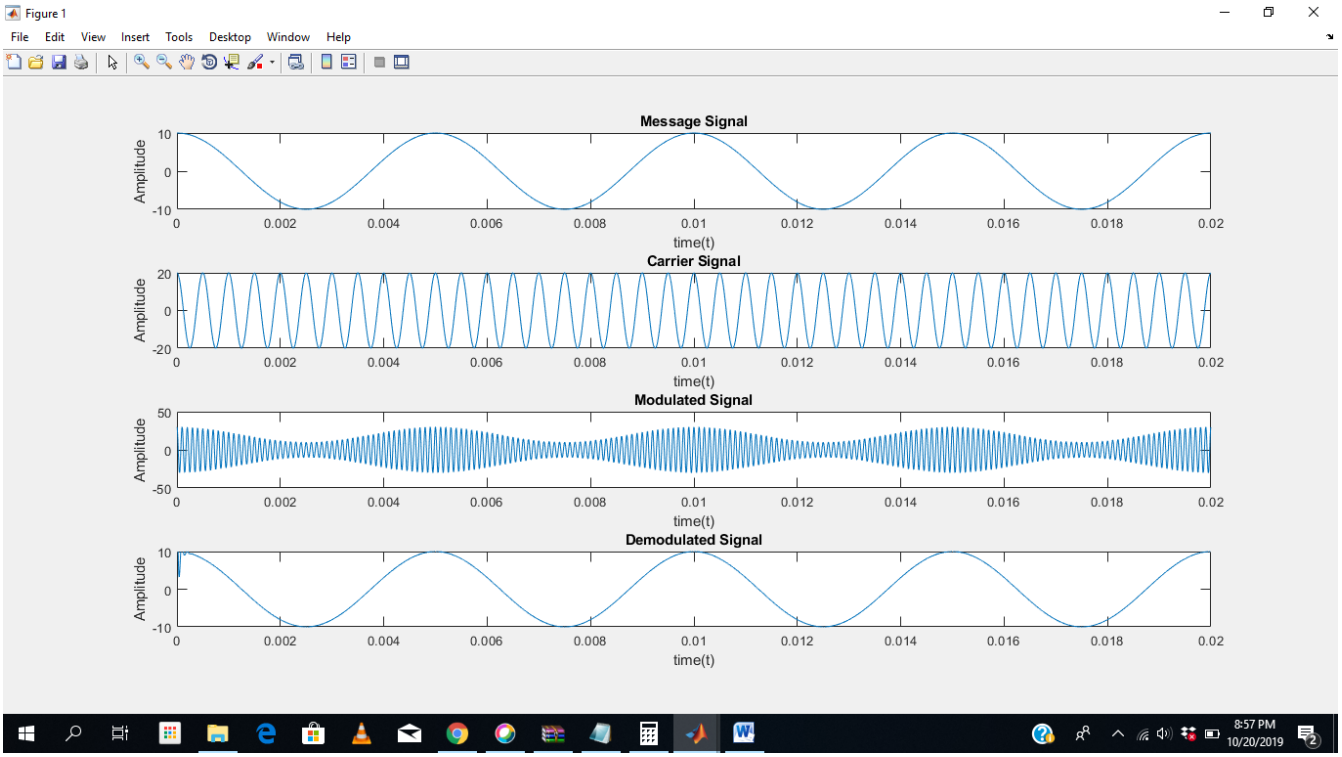
1. Create and plot (both time and frequency domain) a message signal with amplitude 10 and frequency 200 Hz
2. Create and plot (both time and frequency domain) a Carrier signal with amplitude 10/Modulation Index and frequency 2000 Hz
3. Modulate the message signal with the carrier using the desired Modulation Index. Plot modulated signal in both time and frequency domain. Observe/Analyze the output.

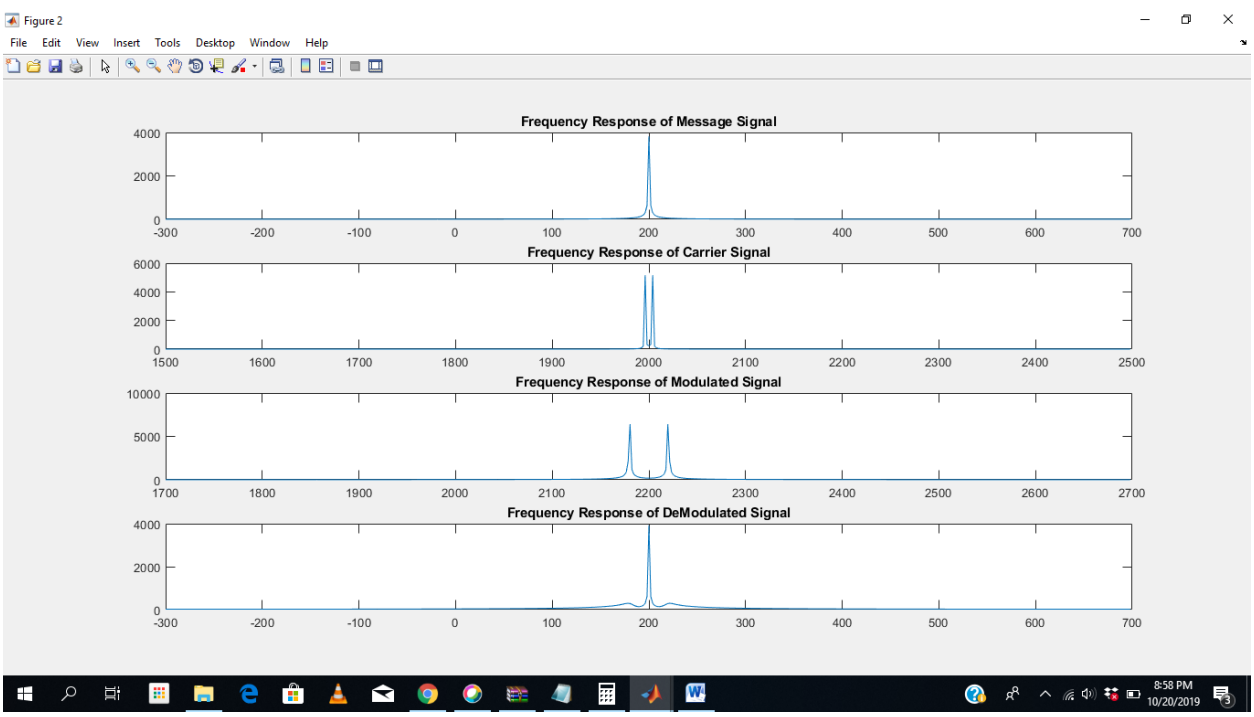
Hint: `y = ammod(ym, fc, 100000, 0, Ac);`

4. Demodulate the Modulated signal . Observe/Analyze the output.

Hint: `z = amdemod(y, fc, 100000, 0, Ac);`

5. Following are the plots of the desired outputs





MATLAB Code:

```
% Matlab Code
Fs = 100000;
t = 0:1/Fs:0.01;
Am = 10;
fm = 200;
fc = 2000;
m = 0.8;
Ac = Am / m;

message_signal = Am * cos(2 * pi * fm * t);
carrier_signal = Ac * cos(2 * pi * fc * t);
modulated_signal = ammod(message_signal, fc, Fs, 0, Ac);
demodulated_signal = amdemod(modulated_signal, fc, Fs, 0, Ac);

freq_axis = linspace(-Fs/2, Fs/2, length(t));

fft_message = abs(fftshift(fft(message_signal)));
fft_carrier = abs(fftshift(fft(carrier_signal)));
fft_modulated = abs(fftshift(fft(modulated_signal)));
fft_demodulated = abs(fftshift(fft(demodulated_signal)));

figure;
```

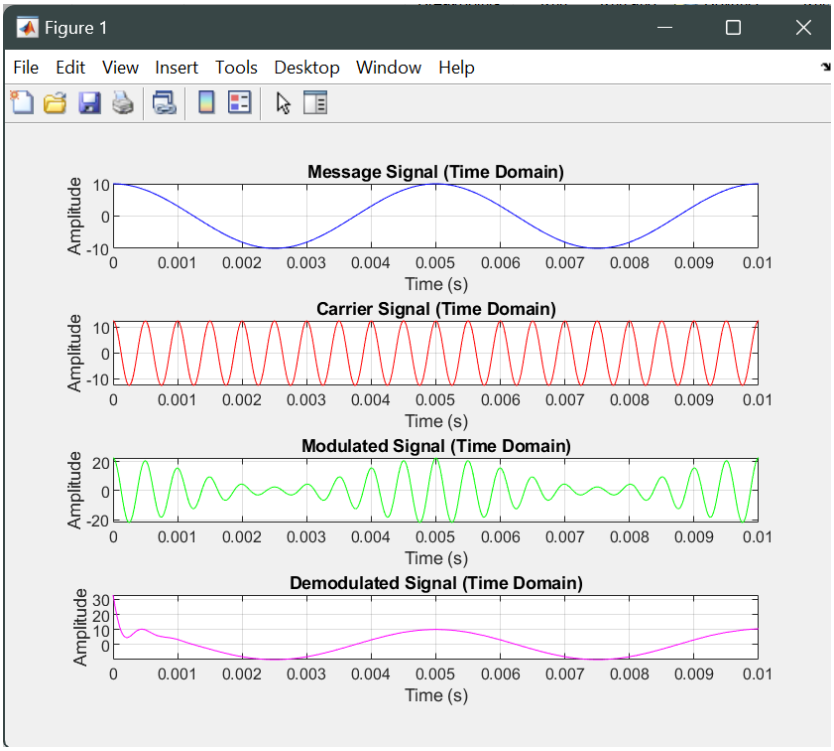
```
figure;
subplot(4, 1, 1);
plot(t, message_signal, 'b');
title('Message Signal (Time Domain)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4, 1, 2);
plot(t, carrier_signal, 'r');
title('Carrier Signal (Time Domain)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4, 1, 3);
plot(t, modulated_signal, 'g');
title('Modulated Signal (Time Domain)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```

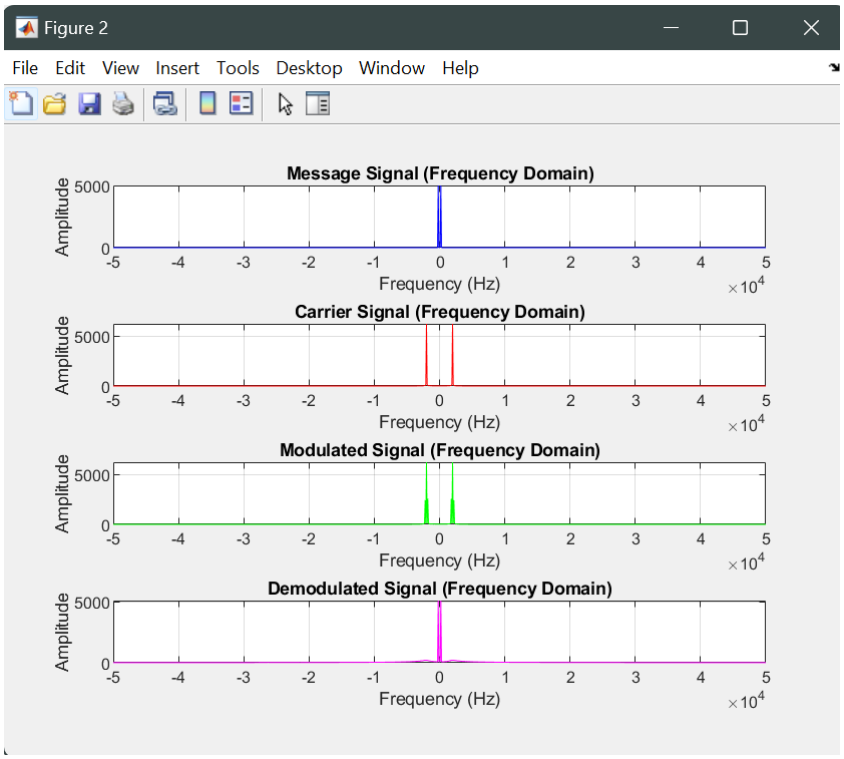
```
subplot(4, 1, 4);
plot(t, demodulated_signal, 'm');
title('Demodulated Signal (Time Domain)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

figure;
subplot(4, 1, 1);
plot(freq_axis, fft_message, 'b');
title('Message Signal (Frequency Domain)');
xlabel('Frequency (Hz)');
ylabel('Amplitude');
grid on;

subplot(4, 1, 2);
plot(freq_axis, fft_carrier, 'r');
title('Carrier Signal (Frequency Domain)');
xlabel('Frequency (Hz)');
ylabel('Amplitude');
grid on;
```

```
subplot(4, 1, 3);
plot(freq_axis, fft_modulated, 'g');
title('Modulated Signal (Frequency Domain)');
xlabel('Frequency (Hz)');
ylabel('Amplitude');
grid on;
```

```
subplot(4, 1, 4);
plot(freq_axis, fft_demodulated, 'm');
title('Demodulated Signal (Frequency Domain)');
xlabel('Frequency (Hz)');
ylabel('Amplitude');
grid on;
```

Output:





Observations:

- The message signal is recovered from the demodulated signal with minimal distortion.
- The frequency domain plots show clear peaks at the message and carrier frequencies.

Analysis of ASK, FSK, PSK, and QPSK

1. **Amplitude Shift Keying (ASK):**
   ASK is a digital modulation technique where the amplitude of the carrier signal is varied according to the digital data.

2. **Frequency Shift Keying (FSK):**
   FSK is a modulation technique where the frequency of the carrier signal is varied in accordance with the digital data.

3. **Phase Shift Keying (PSK):**
   PSK is a digital modulation technique where the phase of the carrier signal is varied in relation to the digital data.

4. **Quadrature Phase Shift Keying (QPSK):**
   QPSK is a type of PSK where the carrier signal is modulated in four distinct phase states, effectively transmitting two bits per symbol.

Tasks:

- Generate a random binary message signal.
- Modulate the message using ASK, FSK, PSK, and QPSK techniques.
- Demodulate each signal and compare it with the original message signal.
- Plot both time and frequency domain representations of the modulated and demodulated signals for each technique.

Amplitude Shift Keying (ASK):

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task1.m | task2.m* | task3.m | task4.m | task5.m

```
Fs = 100000;
Am = 10;
fc = 2000;
message_bits = [1 0 1 1 0 1]; % Example binary message
bit_duration = 0.002; % Duration of one bit
t = 0:1/Fs:bit_duration * length(message_bits) - 1/Fs; % Time vector for all bits
samples_per_bit = Fs * bit_duration; % Ensure this is an integer
message_signal = repelem(message_bits, samples_per_bit);
carrier_signal = Am * cos(2 * pi * fc * t);
modulated_signal = message_signal .* carrier_signal;
demodulated_signal = modulated_signal .* carrier_signal; % Coherent demodulation
demodulated_signal = lowpass(demodulated_signal, 300, Fs); % Low-pass filter

freq_axis = linspace(-Fs/2, Fs/2, length(t));
fft_message = abs(fftshift(fft(message_signal)));
fft_carrier = abs(fftshift(fft(carrier_signal)));
fft_modulated = abs(fftshift(fft(modulated_signal)));
fft_demodulated = abs(fftshift(fft(demodulated_signal)));

figure;
subplot(4, 1, 1);
plot(t, message_signal); title('Message Signal (Time Domain)');
xlabel('Time (s)'); ylabel('Amplitude'); grid on;
```

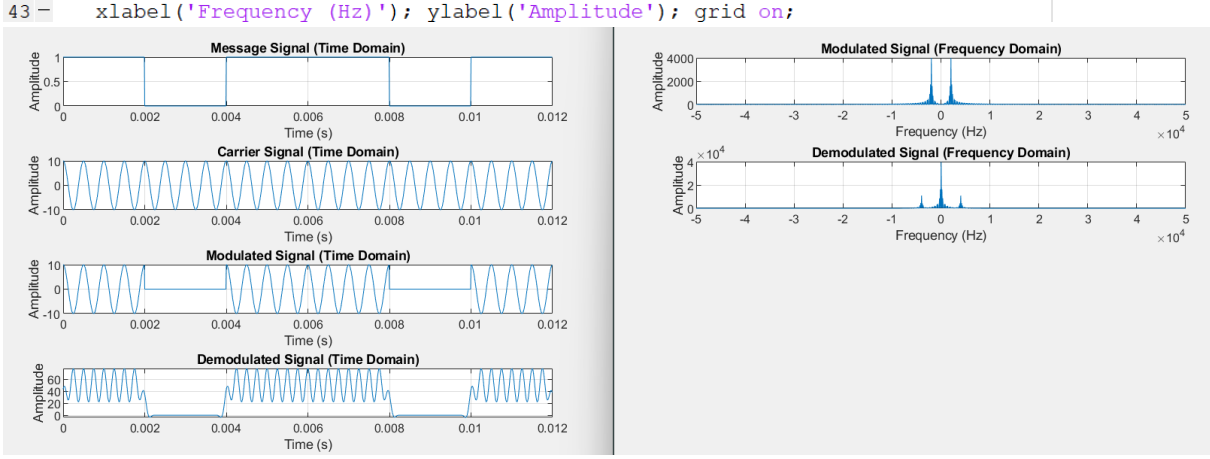
```
subplot(4, 1, 2);
plot(t, carrier_signal); title('Carrier Signal (Time Domain)');
xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(4, 1, 3);
plot(t, modulated_signal); title('Modulated Signal (Time Domain)');
xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(4, 1, 4);
plot(t, demodulated_signal); title('Demodulated Signal (Time Domain)');
xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(4, 1, 3);
plot(freq_axis, fft_modulated); title('Modulated Signal (Frequency Domain)');
xlabel('Frequency (Hz)'); ylabel('Amplitude'); grid on;

subplot(4, 1, 4);
plot(freq_axis, fft_demodulated); title('Demodulated Signal (Frequency Domain)');
xlabel('Frequency (Hz)'); ylabel('Amplitude'); grid on;
```



Frequency Shift Keying (FSK):

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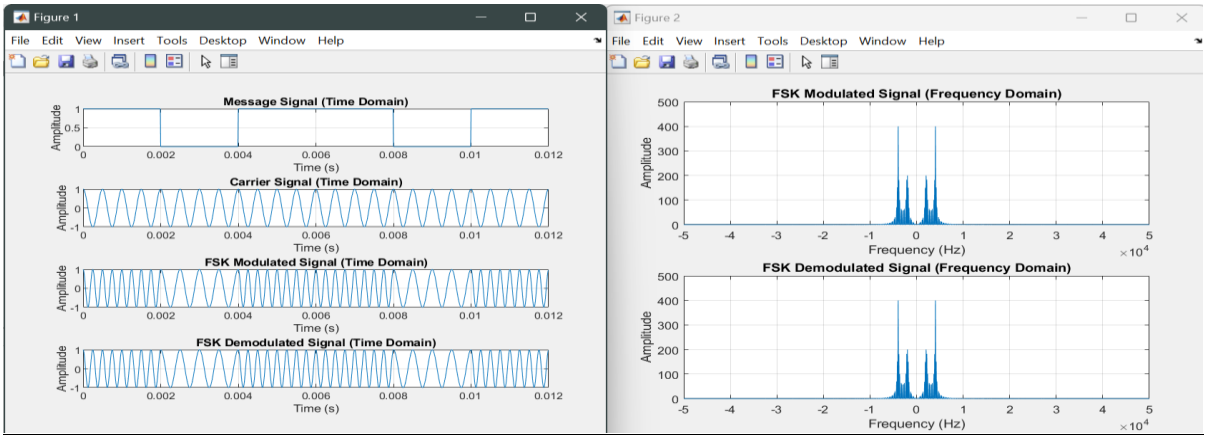
task1.m | task2.m | task3.m* | task4.m | task5.m

```
Fs = 100000;  fc1 = 2000;  fc2 = 4000;
bit_duration = 0.002;  message_bits = [1 0 1 1 0 1];
t = 0:1/Fs:bit_duration * length(message_bits) - 1/Fs; % Time vector for all bits
samples_per_bit = Fs * bit_duration;
message_signal = repelem(message_bits, samples_per_bit);
carrier_signal = cos(2 * pi * fc1 * t); % Carrier for 0 bit
fsk_signal = zeros(1, length(t));
for i = 1:length(message_bits)
    if message_bits(i) == 1
        fsk_signal((i-1)*samples_per_bit+1:i*samples_per_bit) = cos(2*pi*fc2*t((i-1)*sampl
    else
        fsk_signal((i-1)*samples_per_bit+1:i*samples_per_bit) = cos(2*pi*fc1*t((i-1)*sampl
    end
end
fsk_demodulated_signal = zeros(1, length(t));
for i = 1:length(message_bits)
    if message_bits(i) == 1
        fsk_demodulated_signal((i-1)*samples_per_bit+1:i*samples_per_bit) = cos(2*pi*fc2*t
    else
        fsk_demodulated_signal((i-1)*samples_per_bit+1:i*samples_per_bit) = cos(2*pi*fc1*t
    end
end
```

```
freq_axis = linspace(-Fs/2, Fs/2, length(t));
fft_fsk_signal = abs(fftshift(fft(fsk_signal)));
fft_fsk_demodulated = abs(fftshift(fft(fsk_demodulated_signal)));

figure;subplot(4, 1, 1);
plot(t, message_signal);title('Message Signal (Time Domain)');
xlabel('Time (s)');ylabel('Amplitude'); grid on;
subplot(4, 1, 2);
plot(t, carrier_signal);title('Carrier Signal (Time Domain)');
xlabel('Time (s)');ylabel('Amplitude');grid on;
subplot(4, 1, 3);
plot(t, fsk_signal);title('FSK Modulated Signal (Time Domain)');
xlabel('Time (s)');ylabel('Amplitude');grid on;
subplot(4, 1, 4);
plot(t, fsk_demodulated_signal);title('FSK Demodulated Signal (Time Domain)');
xlabel('Time (s)');ylabel('Amplitude');grid on;
figure;subplot(2, 1, 1);
plot(freq_axis, fft_fsk_signal);title('FSK Modulated Signal (Frequency Domain)');
xlabel('Frequency (Hz)');ylabel('Amplitude');grid on;
subplot(2, 1, 2);
plot(freq_axis, fft_fsk_demodulated);title('FSK Demodulated Signal (Frequency Domain)');
xlabel('Frequency (Hz)');ylabel('Amplitude');grid on;
```



Phase Shift Keying (PSK):

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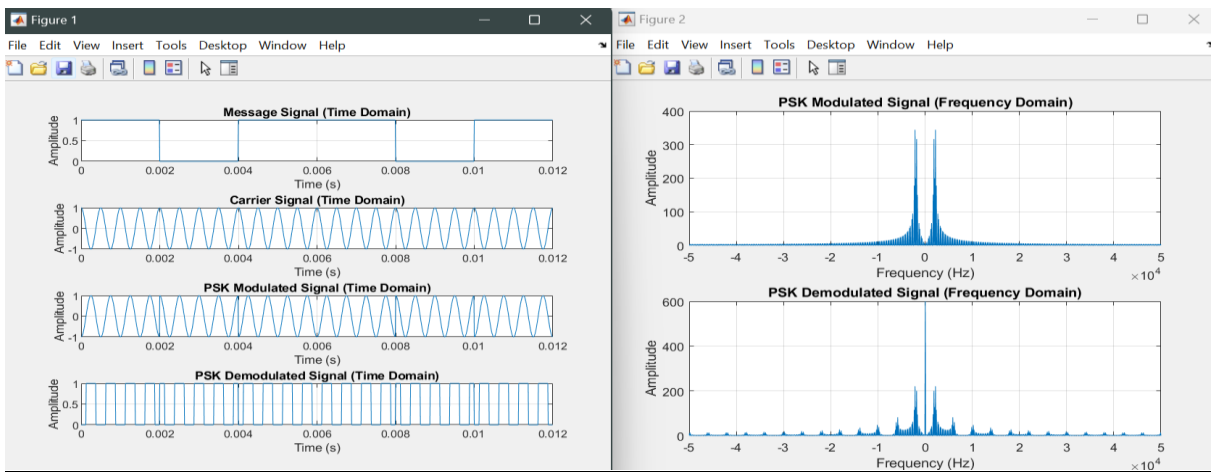
task1.m | task2.m | task3.m | task4.m* | task5.m

```
Fs = 100000;
fc = 2000; % Carrier frequency
bit_duration = 0.002; % Duration of one bit
message_bits = [1 0 1 1 0 1];
t = 0:1/Fs:bit_duration * length(message_bits) - 1/Fs; % Time vector for all bits
samples_per_bit = Fs * bit_duration;
message_signal = repelem(message_bits, samples_per_bit);
carrier_signal = cos(2 * pi * fc * t);
psk_signal = cos(2 * pi * fc * t + pi * message_signal);
psk_demodulated_signal = cos(2 * pi * fc * t + pi * message_signal);
psk_demodulated_signal = psk_demodulated_signal > 0; % Decision threshold at 0
freq_axis = linspace(-Fs/2, Fs/2, length(t));
fft_psk_signal = abs(fftshift(fft(psk_signal)));
fft_psk_demodulated = abs(fftshift(fft(psk_demodulated_signal)));

figure;subplot(4, 1, 1);
plot(t, message_signal);title('Message Signal (Time Domain)');
xlabel('Time (s)');ylabel('Amplitude');grid on;

subplot(4, 1, 2);
plot(t, carrier_signal);title('Carrier Signal (Time Domain)');
xlabel('Time (s)');ylabel('Amplitude');grid on;
```

```
subplot(4, 1, 3);
plot(t, psk_signal);title('PSK Modulated Signal (Time Domain)');
xlabel('Time (s)');ylabel('Amplitude');grid on;

subplot(4, 1, 4);
plot(t, psk_demodulated_signal);title('PSK Demodulated Signal (Time Domain)');
xlabel('Time (s)');ylabel('Amplitude');grid on;

% Frequency-domain plots
figure;subplot(2, 1, 1);
plot(freq_axis, fft_psk_signal);title('PSK Modulated Signal (Frequency Domain)');
xlabel('Frequency (Hz)');ylabel('Amplitude');grid on;

subplot(2, 1, 2);
plot(freq_axis, fft_psk_demodulated);title('PSK Demodulated Signal (Frequency Domain)');
xlabel('Frequency (Hz)');ylabel('Amplitude');grid on;
```



Quadrature Phase Shift Keying (QPSK):

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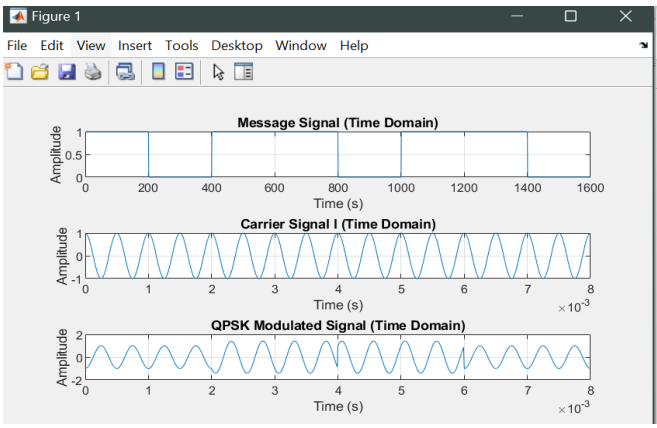
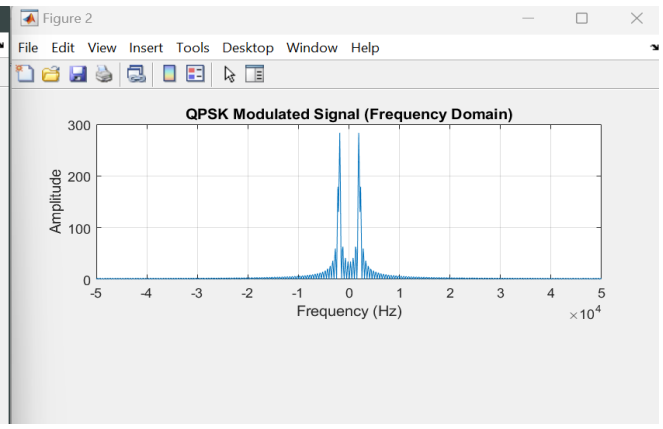
task1.m | task2.m | task3.m | task4.m | task5.m

```
Fs = 100000;
fc = 2000;
bit_duration = 0.002;
message_bits = [1 0 1 1 0 1 1 0];
num_symbols = length(message_bits) / 2;
t = 0:1/Fs:bit_duration * num_symbols - 1/Fs;
symbols = reshape(message_bits, 2, [])';
carrier_I = cos(2*pi*fc*t);
carrier_Q = sin(2*pi*fc*t);
qpsk_signal = zeros(1, length(t));
for i = 1:num_symbols
    if symbols(i, 1) == 0 && symbols(i, 2) == 0
        qpsk_signal((i-1)*Fs*bit_duration+1:i*Fs*bit_duration) = carrier_I((i-1)*Fs*bit_du
    elseif symbols(i, 1) == 0 && symbols(i, 2) == 1
        qpsk_signal((i-1)*Fs*bit_duration+1:i*Fs*bit_duration) = carrier_I((i-1)*Fs*bit_du
    elseif symbols(i, 1) == 1 && symbols(i, 2) == 0
        qpsk_signal((i-1)*Fs*bit_duration+1:i*Fs*bit_duration) = -carrier_I((i-1)*Fs*bit_d
    else
        qpsk_signal((i-1)*Fs*bit_duration+1:i*Fs*bit_duration) = -carrier_I((i-1)*Fs*bit_d
    end
end
freq_axis = linspace(-Fs/2, Fs/2, length(t));
```

```
fft_qpsk_signal = abs(fftshift(fft(qpsk_signal)));

figure; subplot(4, 1, 1);
message_signal = repelem(message_bits, length(t) / num_symbols); % Create message signal f
plot(message_signal); title('Message Signal (Time Domain)');
xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(4, 1, 2);
plot(t, carrier_I); title('Carrier Signal I (Time Domain)');
xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(4, 1, 3);
plot(t, qpsk_signal); title('QPSK Modulated Signal (Time Domain)');
xlabel('Time (s)'); ylabel('Amplitude'); grid on;

figure; subplot(2, 1, 1);
plot(freq_axis, fft_qpsk_signal); title('QPSK Modulated Signal (Frequency Domain)');
xlabel('Frequency (Hz)'); ylabel('Amplitude'); grid on;
```